

TEZPUR UNIVERSITY
Mid Semester Examination, Spring 2023
CO 314 System Software and Compiler Design

Marks : 40

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual question.

1. Write regular expressions for the following tokens of C language programs-
a keyword, an identifier, an integer. [3]
 2. Why is intermediate code created in a two-pass assembler? [2]
 3. Give an example of a macro definition in a typical assembly language, with at least two parameters. Also, give an example of a call of the macro, and what would be the result of pre-processing of that macro call. [2 + 2 + 2]
 4. (a) Compare the advantages and disadvantages of macros and subroutines. [4]
- OR
- (b) What are positional and keyword parameters in a macro? How can default values be specified for macro parameters? [2 + 2]
 5. (a) What is program relocation? What should the relocation table of a program contain? [2 + 2]
 - (b) If a machine language program is created with assumed starting address 0, but it is loaded in a memory region starting from address 2000, what relocation factor should be added to an address value 150 in the program? [2]
 6. What is program linking? What information should the assembler (or compiler) provide to facilitate program linking? [3 + 3]
 7. How does dynamic linking help avoid duplication of executable modules in the memory? [4]
 8. What is meant by precedence of operators in an arithmetic expression? How is a stack useful in evaluating an arithmetic expression? [3 + 3]
 9. (a) What are the precedence relations? What is the precedence relation between '*' and '+'? [2 + 1]
- OR
- (b) How does storing operator precedence relations in a table data structure help? [3]